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Thrust Misalignment Response

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Introduction

There are many possible internal misalignments of rocket structure arising from errors in fabrication and assembly. Most of these cause aerodynamically mediated pitch/yaw torques, and hence have modest impact at low speeds. However, those structural imperfections causing the thrust vector to not pass through the center of mass will cause significant torques even at low speeds. Large low speed torques cause large angular rotations of both body axes and velocity vector, and hence large structural loading and trajectory dispersions.

In general, the response to a structural misalignment can be viewed from two perspectives, global and local. Global response affects the entire airframe. For example, the angle of attack resulting from a thrust misalignment leads to aerodynamic loading on all airframe elements capable of carrying lift.

There are two main applications for the misalignment responses, trajectory dispersions and global structural loading. For both applications, thrust misalignment is, based on experience, the dominant body-fixed structural error. Global structural loading is primarily dependent on angle of attack while trajectory dispersions depend on inertial flight path rotations. One of the main objectives of this analysis is supporting the establishment of good values for the roll rate.

Broadly speaking, this is a problem requiring simultaneous integration of the equations of motion describing all 6 rigid body degrees of freedom. Since Newton's laws are of order two, this composite set of equations is of order 12. Computer programs that numerically integrate this set are customarily called 6 DOF (degrees of freedom). Here, we seek approximate analytical solutions in closed form. This provides not only insight into the effects of the important system variables, it also is far less expensive to code, debug and generate input data. Generally, this memo follows and extends the thrust misalignment approach described in Ref. 1.

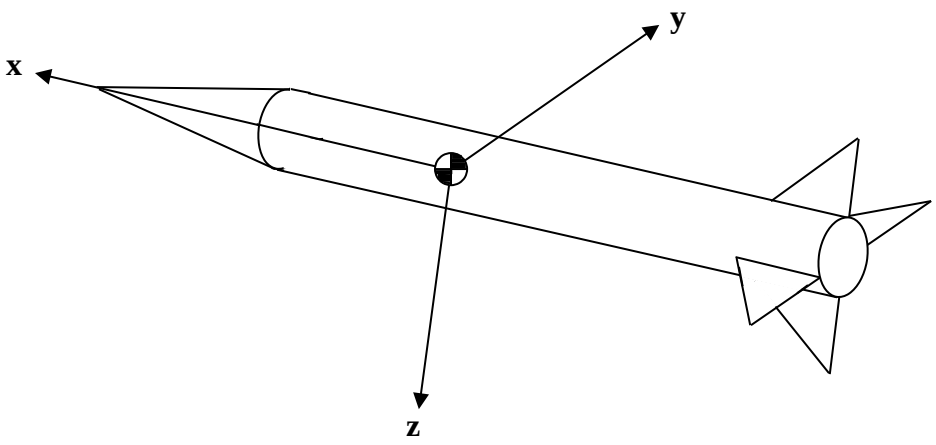


Figure 1 Axes

Nomenclature

Mnemonic	Definition
i	$\sqrt{-1}$
x, y, z	Body fixed axes, x (roll) along the axis of symmetry, y and z (pitch & yaw) forming an orthogonal triad
t	Time, sec
h	Altitude, ft
a	Axial acceleration, feet/sec^2
U	Axial (x axis) velocity, ft/sec
vy, **vz**	Velocity in the y and z directions, ft/sec
ρ	Atmospheric mass density, sl/ft^3
q	Dynamic pressure, lb/ft^2 = ½ **ρU^2**
S	Aerodynamic reference area, ft^2 = **π d^2 /4**
d	Aerodynamic reference length = body diameter, ft
$\mathbf{C_{M\alpha}}$	Pitch moment coefficient derivative with respect to angle of attack, rad^{-1}
$\mathbf{C_{Lp}}$	Roll moment coefficient derivative with respect to **ωx d / 2U**, rad^{-1}
$\mathbf{C_{L\delta F}}$	Roll moment coefficient derivative with respect to the cant angle of a single fin panel, rad^{-1}
N	Number of fin panels
$\boldsymbol{\lambda_P}$	Pitch/yaw wave number, rad/ft

λ_R	Roll angle wave number, rad/ft
I_R	Moment of inertia about the **x**, or roll, axis, sl-ft^2
I_P	Moment of inertia about the pitch or yaw axis, sl-ft^2
Mj	Moment about the jth axis, j = **x, y, z**
ωx, ωy, ωz	Angular rates about the roll, pitch and yaw axes, rad/sec
φ	Roll angle, rad
W	Complex angular rate = **ωy + i ωz**
β, α	Angles of attack and sideslip, rad = **vy/ U**, **vz/U**
A	Complex angle of attack = **β + i α**

V C

omplex lateral velocity = **vy + ivz**

γ	Flight path angle between the local vertical and the velocity vector, rad
γ_D	Dispersive flight path angle, rad
T	Thrust, lb
ℓ	Distance from nozzle throat to rocket center of mass, ft
δ_T	Thrust vector misalignment angle, rad
δ_F	Mean **c**ant angle of a single fin panel, rad
Ω	Lateral c.g. offset from the vehicle center line, ft
L_A	Aerodynamic lift force, lb
$C_{L\alpha}$	Aerodynamic lift coefficient derivative with respect to angle of attack, rad^{-1}
L	Launcher length, ft
E, N	Easterly and northerly components of flight path angle, rad
$( )_o$	Value of ( ) at the time of separation from the launcher
$( )_{BO}$	Value of ( ) at burnout

Note that the right hand rule is the sign convention used for moments and angular rates.

Assumptions

This is a highly simplified analysis. The key assumptions are:

1. The rocket is a rigid body.
2. Furthermore, changes in rocket mass, air density, center of gravity shifts, and Mach No. driven variations in stability derivatives can be neglected.
3. The rocket has pitch-yaw symmetry. That is, all inertial and aerodynamic characteristics are the same about both axes.

4. The rocket's short period damping, while positive, almost vanishes. That is, terms in $\mathbf{C_{Mq}}$, $\mathbf{C_{Z\alpha}}$, $\mathbf{C_{Nr}}$,$\mathbf{C_{Y\beta}}$ and jet damping may be neglected.
5. The only aerodynamic pitch/yaw moment is that due to static stability ($\mathbf{C_{M\alpha}}$ & $\mathbf{C_{N\beta}}$).
6. The axial (**x** axis) translational motion may be approximated by constant acceleration.
7. The rocket's roll moment of inertia is negligibly small compared to the pitch/yaw moment of inertia, and to the aerodynamic roll driving and damping moments.
8. The nominal trajectory is vertical.
9. The thrust misalignment applies a torque about the pitch axis, but because of assumption 2., this has arbitrary phasing.
10. $\boldsymbol{\omega y, \omega z}$, α, β, δ and Ω are all small compared to unity. This is a linearized analysis.
11. The component of gravity along the **y** and **z** axes may be neglected when estimating the angle of attack because this is a short period analysis.
12. The lateral force component of the misaligned thrust is neglected because the thrust misalignment torque is much more important.

The ultimate rationale for adopting such a severely simplified dynamics model is that the thrust misalignment response is proportional to the δ and Ω perturbation errors, and these are known but poorly. Reference 4 provides data for these parameters, but it is "best practice" data valid in an industrial factory environment where rockets were built with hard tooling and a formal Quality Assurance program in place. There's no great value in using a highly accurate dynamics model if the result is necessarily corrupted by inaccurate perturbations.

Analysis

First, this is a linear analysis…sounding rockets are very dynamically linear with a few stark, shocking exceptions such as roll lock in. Second, the linear differential equations we must integrate have complex coefficients as a consequence of the pitch-yaw symmetry possessed by such machines. Third, the coefficients are profoundly time-varying. Fortunately, these apparently scary aspects can be successfully addressed. The key trick is to change the independent variable from time to distance along the flight path, which if vertical flight is assumed, amounts to altitude.

Let's begin with the assumed solution per Assumption 6. Then,

$$\mathbf{U = a\,t = \sqrt{2\,a\,h}}. \tag{1}$$

Since we are going to change independent variables, this is a good place to note that the chain rule gives (vertical flight per Assumption 8.)

$$\mathbf{d/dt = d/dh\ dh/dt = U\,d/dh}. \tag{2}$$

Also, the roll rate is (per Assumption 7.) proportional to axial velocity[1]:

$$\omega x = \lambda_R U, \tag{3}$$

where λ_R is a constant of proportionality dependent on the fin cant angles. Integrating once more to find the roll angle is easy:

$$d\varphi/dt = \omega x = \lambda_R U = U\, d\varphi/dh, \text{ or}$$

$$\varphi = \lambda_R (h. - L) \tag{4}$$

From this it can be seen that λ_R has the characteristics of a wave number.

Newton's Second law written for axes fixed to a rigid body can be found in any good dynamics text such as Ref. (2). They are

$$\begin{aligned} My &= I_P\, d\omega y/dt - (I_P - I_R)\, \omega x\, \omega z, \\ Mz &= I_P\, d\omega z/dt + (I_P - I_R)\, \omega x\, \omega y, \\ Fy/m &= dvy/dt - \omega x\, vz + U\, \omega z = 0, \text{ and} \\ Fz/m &= dvz/dt + \omega x\, vy - U\, \omega y = 0. \end{aligned} \tag{5}$$

Next, the external moments are given by

$$My = q\, S\, d\, C_{M\alpha}\, \alpha + T\,(\ell\, \delta_T + \Omega) + q\, S\, d\, C_{M\delta F}\, \delta_F,$$

and

$$Mz = - q\, S\, d\, C_{M\alpha}\, \beta, \tag{6}$$

because the sign conventions used and Assumption 3 imply that $C_{M\alpha} = - C_{N\beta}$. The perturbing torques, $T\,(\ell\, \delta_T + \Omega)$, are usually random errors. Experimental data for these can be found in Ref. 3.

Use the customary definition of the angles of attack and sideslip:

$$\alpha \equiv vz / U, \text{ and } \beta \equiv vy / U,$$

and the definition of dynamic pressure, $q = \frac{1}{2} \rho U^2$, to find that

$$My = \tfrac{1}{2}\, \rho\, S\, d\, C_{M\alpha}\, vz\, U + T\,(\ell\, \delta_T + \Omega) + \tfrac{1}{2}\, \rho\, S\, d\, C_{M\delta F}\, \delta_F\, U^2, \text{ and} \tag{7}$$

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[1] Since the roll moment of inertia is negligibly small, the roll moment equation becomes $C_{Lp}\,(\omega x\, d/2\, U) + C_{L\delta F}\, \Sigma\, \delta_F = 0$. Therefore, $\lambda_R = -2\, N\, C_{L\delta F}\, \Sigma\, \delta_F / C_{Lp}\, d$.

$$\mathbf{Mz} = -\tfrac{1}{2}\,\rho\,\mathbf{S\,d}\,C_{M\alpha}\,\mathbf{vy\,U.}$$

Before substituting eq's. (7) back into eq's. (5) note that there are two terms in the thrust "misalignment" torque. The real misalignment torque comes from failure to align the thrust chamber/nozzle parallel to the rocket's symmetry axis. The c.g. offset term comes from a failure to mass balance the rocket. These two terms are statistically independent, and eventually must be treated as such.

Then, invoking Assumption 7,

$$\begin{aligned} I_P\,d\omega y/dt - I_P\,\lambda_R\,U\,\omega z &= \tfrac{1}{2}\,\rho\,S\,d\,C_{M\alpha}\,vz\,U + T\,(\ell\,\delta_T + \Omega), \\ I_P\,d\omega z/dt + I_P\,\lambda_R\,U\,\omega y &= -\tfrac{1}{2}\,\rho\,S\,d\,C_{M\alpha}\,vy\,U, \\ dvy/dt - \lambda_R\,U\,vz + U\,\omega z &= 0, \text{ and} \\ dvz/dt + \lambda_R\,U\,vy - U\,\omega y &= 0. \end{aligned} \tag{8}$$

Now, the math becomes interesting. First, change the independent variable from time to altitude:

$$\begin{aligned} \tfrac{1}{2}\,\rho\,S\,d\,C_{M\alpha}\,vz + T\,(\ell\,\delta_T + \Omega)/U &= I_P\,d\,\omega y/dh - I_P\,\lambda_R\,\omega z, \\ -\tfrac{1}{2}\,\rho\,S\,d\,C_{M\alpha}\,vy &= I_P\,d\,\omega z/dh + I_P\,\lambda_R\,\omega y, \\ dvy/dh - \lambda_R\,vz + \omega z &= 0, \text{ and} \\ dvz/dh + \lambda_R\,vy - \omega y &= 0. \end{aligned} \tag{9}$$

Now, define

$$\lambda_P^2 = -\tfrac{1}{2}\,\rho\,S\,d\,C_{M\alpha}/I_P \tag{10}$$

Here λ_P is the pitch / yaw wave number in radians per foot.
With this, we have

$$\begin{aligned} -\lambda_P^2\,vz + T\,(\ell\,\delta_T + \Omega)/I_P\,U &= d\,\omega y/dh - \lambda_R\,\omega z, \\ \lambda_P^2\,vy &= d\,\omega z/dh + \lambda_R\,\omega y, \\ dvy/dh - \lambda_R\,vz + \omega z &= 0, \text{ and} \\ dvz/dh + \lambda_R\,vy - \omega y &= 0. \end{aligned} \tag{11}$$

The next trick is to exploit rotational symmetry. Define two new complex variables:

$$V = vy + i\,vz, \text{ and}$$

(12)

$$\mathbf{W} = \omega y + \mathbf{i}\ \omega z.$$

Multiply the second and fourth eq's. (8) by $\sqrt{-1} = \mathbf{i}$, and add to the first and third eq's. (8) respectively. The result is

$$\mathbf{d\,W/dh} + \mathbf{i}\,\lambda_R\,\mathbf{W} - \mathbf{i}\,\lambda_P^2\,\mathbf{V} = \mathbf{T}\,(\,\ell\,\delta_T + \Omega\,)\,/\,\mathbf{I_P}\,\mathbf{U}\,, \text{ and}$$

(13)

$$\mathbf{d\,V/dh} + \mathbf{i}\,\lambda_R\,\mathbf{V} - \mathbf{i}\,\mathbf{W} = \mathbf{0}.$$

Eliminating **W** from these gives

$$\mathbf{d^2\,V/dh^2} + 2\,\mathbf{i}\,\lambda_R\,\mathbf{d\,V/dh} + (\,\lambda_P^2 - \lambda_R^2\,)\,\mathbf{V} = \mathbf{i}\,\mathbf{T}\,(\,\ell\,\delta_T + \Omega\,)\,/\,\mathbf{I_P}\,\mathbf{U}$$

(14)

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Steady State Angle of Attack

Before pursuing a dynamical solution, suppose that after the passage of a long time the small damping terms neglected in the derivation so far have caused the transient motion to completely die down. The description below is the extension of Ref.(3) to the case of a symmetrical vehicle accelerating through the atmosphere.

Then, recalling that

$$\mathbf{A} = \mathbf{V/U}, \tag{15}$$

$$(\,\lambda_P^2 - \lambda_R^2\,)\,\mathbf{A} = \mathbf{i}\,\mathbf{T}\,(\,\ell\,\delta_T + \Omega\,)\,/\,\mathbf{I_P}\,\mathbf{U^2}.$$

Noting that the forcing function is entirely imaginary, we can replace **A** with α to give the steady state angle of attack,

$$\alpha_{ss} = \mathbf{T}\,(\,\ell\,\delta_T + \Omega\,)\,/\,\mathbf{I_P}\,\mathbf{U^2}\,(\lambda_P^2 - \lambda_R^2\,) \tag{16}$$

In Fig. 2 below the steady state thrust misalignment response, α_{ss}, normalized to its non-rolling value, is plotted against λ_R/λ_P.

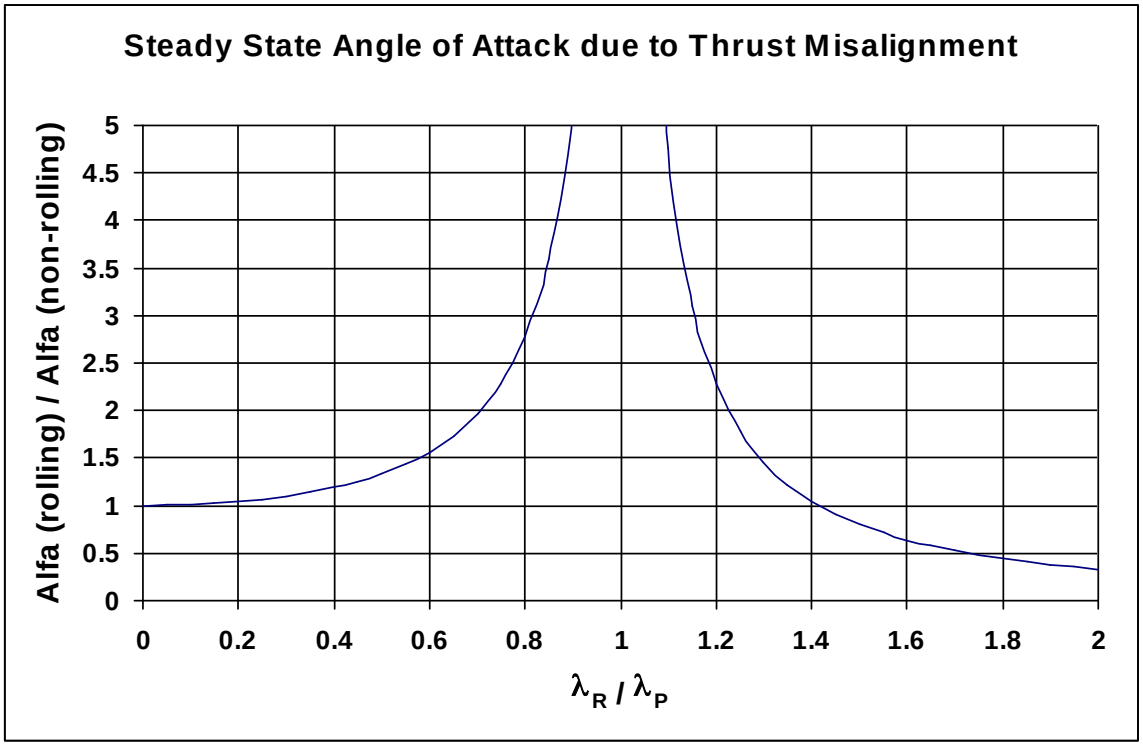


Figure 2 Influence of roll rate on steady state angle of attack due to thrust misalignment

Several ideas should be taken from this graph. First, while the angle of attack appears to become infinite at $\lambda_R/\lambda_P = 1$, in reality the damping terms neglected in the derivation will cap the plot at a finite, but high value. Second, from time to time it occurs to observant people that a roll rate significantly larger than the pitch/yaw natural frequency will reduce both structural loading and trajectory dispersions. Follow this path with care, because if the roll rate matches any of the body bending natural frequencies, an explosive structural failure will occur. Finally, it can easily be shown that all body-fixed misalignment torques, e.g., fin misalignment torques, produce a similar steady state response curve.

The above is a static trim analysis, frequently used in the estimation of structural loading because it's so simple. It is also possible to develop a dynamic analysis from eq. (14). The result is described in terms of Fresnel integrals. While there are formulas for these suitable for coding into software, the extra effort compared to the improvement in accuracy is doubtful.

References

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